

IoT-Enabled Aquaculture System with Feeding, Mortality Detection, Budgeting, Monitoring, and Alerts

Project ID: 25-26J-295

Project Proposal Report

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BSc (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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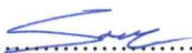
DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for their undergraduate dissertation under my supervision.

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ABSTRACT

This project reveals an IoT aquaculture management system that combines energy metering, expense entry, and analytics. The ESP32 microcontroller at the heart of the hardware samples electricity from the farm's main supply with an ACS712 current sensor and sends the data to the cloud. Instead of measuring electricity for each device, it is measured once for the whole project. This matches the utility bill and makes installation easier. A point-of-sale module that is built into the system keeps track of non-utility spending in daily operations. Costs for power are recorded automatically, but staff members have to enter costs for labour, feed, chemicals, maintenance, and other things using POS forms. Smart plug integration lets you switch and diagnose without changing the price. Using defined cycle windows and stock counts, the platform figures out the cost per fish, cost per kilogram, and cycle based cost. Managers can plan purchases and set budgets with the help of time series models, which predict energy use and total spending. Anomaly checks mark unusual entries or consumption for review. A secure dashboard shows live trends, ledger views, and audit trails, and lets you set role permissions. The focus of implementation is on low cost and dependability. The ESP32 can sense things, connect to secure Wi-Fi, and store data in a buffer when the network goes down. Firebase is used by the backend to sync data, set rules, and send alerts. You can export cycle reports and ledgers as PDFs and Excel files to share and keep them. The result is a single source of truth that connects farm events to their costs and helps with accurate benchmarking and tighter budgets.

Keywords: IOT energy monitoring, Aquaculture cost management, smart plug integration, time-series forecasting, dashboard analytics.

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LIST OF ABBREVIATIONS

Abbreviation	Description
IoT	Internet of Things
TLS	Transport Layer Security
ISO	International Organization for Standardization
ESP 32	Espressif 32-bit microcontroller
KPI	Key Performance Indicator
POS	Point Of Sale
FCM	Firebase Cloud Messaging
SoC	State of Charge
ROI	Return on Investment

1 INTRODUCTION

1.1 Background study

Aquaculture businesses use pumps, aerators, and sometimes temperature control systems that run on electricity for long periods of time to keep fish healthy and water clean. Electricity costs money along with feed, fingerlings, chemicals, labour, maintenance, and overhead because these loads often happen during peak tariff windows and seasonal demands. For small and medium-sized farms with tight budgets, broken records and late bills make it hard to see where money goes, make it hard to compare cycles fairly, and make it hard to see how changes like longer aeration during hot spells will affect the budget [1].

Normal practices only show part of the picture. Monthly bills add up usage from different times and activities, making it hard to see when costs go up and why. You can use spreadsheets and notebooks to keep track of your purchases, but they're not good for making decisions in real time because entries can be late, categories can be inconsistent, and mistakes happen a lot [2]. Many sites don't have a single point of sale that can keep track of expenses and farm transactions with the same item codes, units, and approvals. Without that structure, managers deal with overruns instead of planning costs ahead of time. They have a hard time figuring out how much it costs to keep a fish alive and don't have any proof to change duty cycles, maintenance intervals, or feeding strategies. People often miss chances to save money, like moving non-critical loads away from peak pricing or finding a broken pump by its unusual draw. If you only look at your monthly bill [3] to see how much energy you use, you can't make fair comparisons across production cycles because you don't have timely, sitewide data.

Recent advances in low-cost sensors and edge computing have made a unified approach possible. An ESP32 can connect to an ACS712 current sensor and a ZMPT101B voltage sensor, or to certified smart meters at the main feeder. The device takes short samples of current and voltage, figures out power and energy, and changes usage into cost under flat or time-of-use tariffs. It's easier to install and matches utility billing to measure electricity once for the whole project instead of for each device. An integrated POS module works with telemetry by letting employees fill out forms to enter labour, feed, chemicals, fingerlings, maintenance, harvest proceeds, and services. Items have codes, units, batch notes, supplier names, unit costs, and roles for approvals

and audit trails. Using both automatic energy records and structured POS logs by pond and production cycle gives you KPIs like cycle-based cost, cost per fish, fixed and variable splits, and budget run rate. Cloud alerts let you know when there are spikes, zero draw failures, or changes to the plan. Time series models turn patterns from the past into short-term cost forecasts that help managers plan purchases and set limits. You can export cycle-based cost reports to PDF and Excel for review and storage. This helps aquaculture that is based on data and is good for the environment.

1.2 Literature survey

Energy consumption in aquaculture

Fish farms use a lot of energy because they need to run pumps, aerators, and sometimes temperature controls for long periods of time to keep the water safe. Reviews show that the energy intensity of different systems and species can be very different because of pumping head, aeration needs, and climate. In reality, many farms don't keep up-to-date, detailed records and instead rely on monthly utility bills that don't show when or why draw increased. That makes it hard to see waste, makes it hard to compare cycles fairly, and makes it harder to judge changes in operations, like changing aeration schedules. The literature emphasizes the significance of higher frequency metering for cost management. For this project, however, electricity is measured at the main feeder instead of by device. Cost attribution is also better because feeder totals are combined with production logs and structured expense entries from an integrated point of sale.

Smart energy monitoring using the Internet of Things

Smart home and smart grid research, other than aquaculture, uses plug-level meters, current sensors like the ACS712, and voltage sensors like the ZMPT101B or certified smart meters to get accurate estimates of power and energy. Most designs send data to an edge controller, which then calculates the kWh and cost based on the tariff and shows appliance dashboards. Numerous studies indicate that straightforward instrumentation coupled with prompt feedback can result in quantifiable savings. In aquaculture, this means always keeping an eye on critical loads and changing draw into cost. In this method, an ESP32 samples the main supply's current and voltage for whole-site accounting, and POS forms keep track of labor, feed, chemicals, and other inputs. Telemetry and POS records work together to show device groups and cycles, the cost per fish, and exceptions without having to wire every asset.

Forecasting costs and energy

Buildings and grids have a long history of short-term energy forecasting. When there isn't much data, classical time-series families like ARIMA, ETS,[4] and related additive models can show daily and weekly seasonality. For small farms, one-day-ahead and next-cycle forecasts of kWh and cost with confidence bands are enough for planning, budgeting, and testing what-if scenarios, like changes in feed or tariff prices. The literature shows that well-tuned, simple models often work well on sparse datasets, which is perfect for lightweight [5]cloud services or edge environments that are limited in space and run alongside ESP32 telemetry.

Safety, dashboards, and cloud integration

Real-time databases and messaging in IoT backends make it possible to store data, let multiple users access it, and send alerts to mobile devices. Dashboards bring together trends, categories, exceptions, and forecasts. They also let you export cycle reports in PDF and Excel for approval and audit. Work on power quality and motor protection shows how important it is to monitor voltage and use interlocks that stop pumps from starting outside of safe ranges to cut down on damage and downtime. These ideas work for small farms that use cheap sensors, relay control, and role-based POS workflows to make sure that financial and electrical records match up. This makes things more open and organized.

1.3 Research gap

Current aquaculture IoT projects focus on fish health and water quality, while energy analytics and financial visibility get less attention. Most farms still get monthly utility bills that add up usage from all over the site and over time. This makes it hard to see the times and practices that cost money. Accounting is often done in spreadsheets or other basic farm tools that don't have telemetry links. Because of this, managers can't reliably combine measured draw, logged spending, and stock counts to make clear, fair cycle-based KPIs like total spend or cost per fish.

Smart energy studies outside of aquaculture use sensors like the ACS712 and ZMPT101B [10] to get accurate readings, send alerts, and make predictions. But these tools don't often link to aquaculture accounting for feed, fingerlings, labour, and chemicals. They don't often use voltage safety measures and don't often calculate cycle KPIs. There are industrial energy systems, but they are expensive and hard to use for small outdoor ponds. A site-level method that measures at the main feeder and combines totals with POS expense logs is easier.

The research gap is a low-cost, integrated module for small farms. Using an ESP32, it measures the electricity at the main supply of the site, converts usage to cost under the tariff, and matches those readings with point-of-sale entries by pond and production cycle. The system figures out KPIs like cost per fish, cycle-based cost, category splits, fixed and variable shares, and more. It makes short-term predictions of energy and total cost with confidence bands, finds unusual events like spikes, zero draw events, and budget run rate drift, and sends out cloud alerts. During mains, voltage interlocks protect pumps. You can export cycle reports as PDF and Excel files. This work offers an economical method for managing energy and finances in small-scale aquaculture.

Yes = ✓

No = ✗

Partial/unclear = ~

Study/year	setting	Device-level energy metering	Expense logging	Forecasting (energy /cost)	Cost complexity	Key limitation & new function
Saha et al./2023(IOT fish-farm water quality)	Aquaculture ponds	✗	✗	✗	Low-Med	Env-monitoring only, new function adds metering, expense logging forecasting for costs
Bujas et al.,2002(Energy in aquaculture - review)	Review (multiple systems)	~	✗	✗	-	Survey without implementation, new function delivers a practical integrated module
Ben Dhaou et al.,2023(Smart-plug energy monitoring)	Smart home	✓	✗	~	Low-Med	Not aquaculture, no cycle KPIs/expenses, new function adds farm expense logging and cycle budgeting.
Pierre et al.,2023 (ARIMA/LSTM peak demand)	Buildings/grid	~	✗	✓	Med	Forecasting only (assumes data) adds farm expense logging & cycle budgeting

Industrial EMS (vendor platform)	Industrial sites	✓	~	✓	High	Over engineered/expensive for small ponds; new function is low-cost and farm specific
Proposed system (this work)	Aquaculture Ponds	✓	✓	✓	Low	Integrated low-cost solution; metering, expense logging, cost forecasting tailored to ponds.

Table 1.1: Comparison chart

1.4 Research problem

Small aquaculture farms can't easily find a cheap way to see how their energy use affects their daily costs, how those costs add up with labour, feed, and chemicals, and what the costs are likely to be in the next cycle. Most smart energy tools are made for homes, not ponds, and they don't often keep track of cycles or key performance indicators (KPIs) like the cost of each fish that survives. Aquaculture IoT projects focus on either the health of the fish or the quality of the water. So managers depend on monthly bills and spreadsheets that are too rough and too late for proactive control. The research problem is to create and test a farm-specific module that uses an ESP32 to measure electricity at the main feeder, track non-utility costs through a point-of-sale workflow by pond and production cycle, and combine these streams into analytics KPIs, time series cost forecasts, and anomaly alerts, all while making sure that pumps are safe from voltage overloads. The solution must provide accurate budgeting, quick alerts, offline resilience, and a footprint that smallholders can keep up with. It should let you export cycle reports to PDF and Excel. This way, you can use timely, reliable evidence to help you make daily decisions about duty cycles, maintenance, and feeding.

2 RESEARCH OBJECTIVE

2.1 Main objective

To develop and use a low-cost, edge-enabled module for small-scale aquaculture that combines site-level energy metering at the main feeder with structured POS expense logging and gives useful cost analytics like key performance indicators (KPIs), time-series forecasts, anomaly alerts, and voltage-safety interlocks. This way, farm managers can plan budgets ahead of time and cut down on waste.

2.2 Specific objective

- Set up energy metering at the site level: connect ZMPT101B or zero-flux voltage sensing and ACS712 current sensing, or a certified smart meter, to an ESP32 at the main feeder to measure real-time power, energy in kWh, and cost. Calibrate with a reference meter, aiming for 5 percent.
- Create a structured way to keep track of costs by giving each pond and production cycle a point-of-sale (POS) interface to record feed, fingerlings, drugs and chemicals, labour, maintenance, and overhead, with categories and notes that can be changed.
- Figure out aquaculture KPIs by looking at total spending, cost per surviving fish, category splits, and fixed vs. variable costs. Make it possible to compare ponds and cycles fairly.
- Give users the ability to run "what if" simulations. For example, they should be able to change important inputs like "tariff plus ten percent" and "feed plus eight percent" and then recalculate KPIs and see how the budget is affected.
- Predict energy use and total cost: use a simple time-series forecaster like ARIMA or ETS as a baseline to confidently guess the next period's kWh and LKR with confidence bands for budgeting.
- Set up alerts and reports: For safety and cost issues, use FCM or SMS to send push notifications. For stakeholders to review, export cycle summaries and dashboards to PDF and Excel.
- Set up a voltage-safety interlock: if the mains voltage goes outside a certain safe range, stop the pump and motor from starting and send alerts right away. Record recovery events that bring the system back to a safe state.
- Use store-and-forward buffering, watchdog restarts, and cloud sync protected by TLS to make sure that the system works safely and reliably even when it's not connected to the internet. Use ISO 8601 to timestamp data in local time and keep only the necessary aggregates.

- Give questions and answers in plain English: let users ask questions about historical drivers like the biggest cost contributors from the last cycle and forecast reasons like the expected cost for next month and why.

3 METHODOLOGY

Overview

This study follows to a build test refine cycle. An ESP32 edge node measures [6] the mains voltage and line current at the main feeder of the farm. A simple POS interface keeps track of costs that aren't related to utilities by pond and production cycle. The edge node calculates power, energy in kWh, and rolling cost, combines these with expense entries, and creates KPIs, alerts for anomalies, forecasts, and reports. A voltage-protection interlock stops the pump from starting if the mains voltage is outside a safe range and sends notifications right away. Short field tests confirm that design choices are accurate, reliable, and fast in outdoor conditions and when the connection is not always there.

Hardware configuration

The edge device is an ESP32 that is protected from water. A ZMPT101B or zero-flux sensor measures AC voltage, and an ACS712 [7] measures line current at the main supply. An isolated ADC path is used when necessary. When you can get one, a certified smart meter gives you a reference channel for calibration. A relay or contactor enable line is driven by the pump or motor interlock. Components are fused and relieved of strain, and qualified people watch over work on the mains. The node gets its power from a regulated DC supply or battery, and it can also get a trickle charge from a solar panel to work on its own in the field.

Safety and calibration

For the first calibration, a reference meter and known loads between 60 and 500 W are used. Two-point or multi-point procedures give you scale and offset constants for voltage and current, as well as an empirical factor k that takes into account the power factor of common inductive loads. Sampling is done slowly, in seconds, with rolling averages and outlier rejection to keep readings steady without hiding real spikes. For example, a safe voltage band of $230\text{ V} \pm 10\%$ [8] controls the interlock. Commands to

start that are outside of this range are blocked, and the operator must confirm recovery after a hard fault.

Data acquisition & local processing

The ESP32 runs a loop that timestamps frames of V and I sample, calculates instantaneous power P and cumulative energy E, and changes usage to cost based on flat or time-of-use tariffs. Measurements are saved to a local store with store-and-forward buffering so that data and alerts stay available even when the network goes down. Pond and cycle IDs, tariff tables, and cost centres linked to POS items are all examples of configurable metadata. Settings stay the same after a restart, and the device clock is used for timekeeping with periodic sync.

Analytics and key performance indicators

Every time the sync happens, the analytics layer updates the total spend, the share of energy, the category splits, the fixed vs. variable allocations, and the cost per fish that survives. We keep an eye on the run-rate and budget all the time. With a "what-if" panel, users can change inputs like "tariff plus ten percent" and "feed plus eight percent," and then recalculate KPIs [9] for planning and approvals. Site-level metering is compared with POS entries to figure out costs by pond and cycle without having to wire every device.

Alerts, reports, and Q&A

FCM push messages, with the option to send SMS, are sent for confirmed events. These messages include the type of event, the time, the site or pond, and a short context. Cool-down windows stop duplicates from happening. Weekly and on-demand PDF and Excel exports include KPIs, trends, outliers, and notes that need to be signed off on. An optional chat-style Q&A takes natural-language prompts like "What were the top cost drivers last cycle?" and turns them into parameterized queries that give short, data-backed answers.

Workflow for operations

When the node starts up, it tests its own sensors, loads settings for tariffs, devices, and safe ranges, and connects to the network. It takes samples of V and I, updates KPIs, checks for voltage safety and problems, and puts alerts in a queue. When the link goes down, data and messages are stored and sent when the service comes back up. If a

service fails, a watchdog restarts it. Logs use local time ISO-8601 timestamps to make it easier for teams and systems to find things.

Evaluation plan

Validation checks that sensors are accurate within five percent of a reference meter for power and energy, that interlock latency is two seconds to block starts outside the safe band, that alert latency is fifteen seconds from event to phone, that initial forecasting MAE is fifteen to twenty-five percent with improvement as data accumulates, that there is a low false-positive rate under debris and idle states, and that there is twenty-four hours of continuous uptime with auto-recovery. Thresholds and model settings are adjusted for small outdoor pond operations until the goals are met.

3.1 System overview diagram

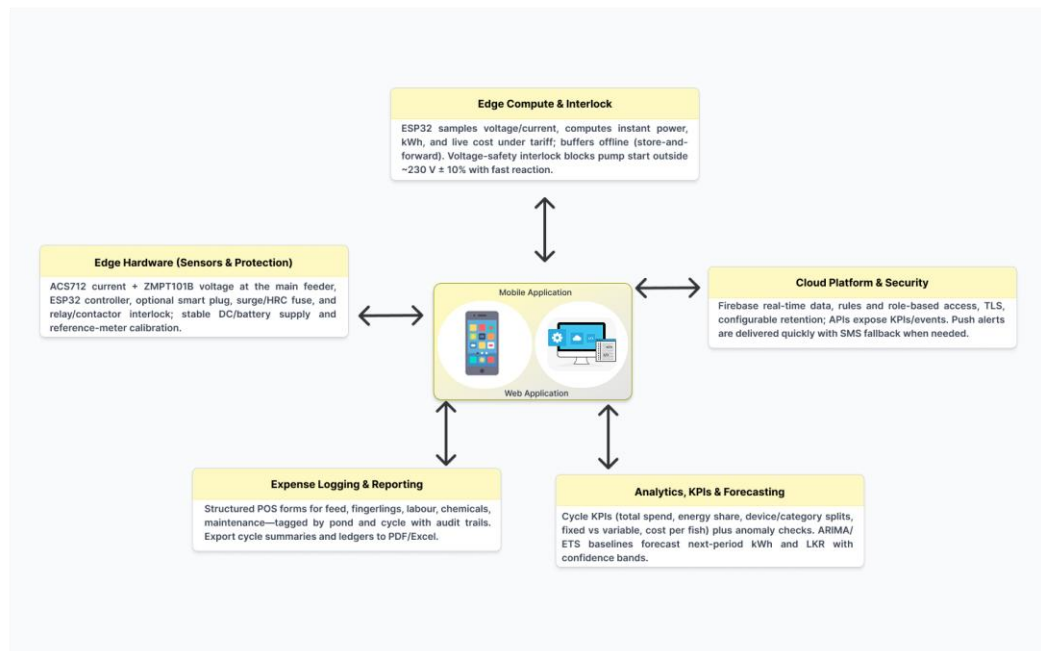


Figure 3.1: System overview

3.2 Component overview diagram

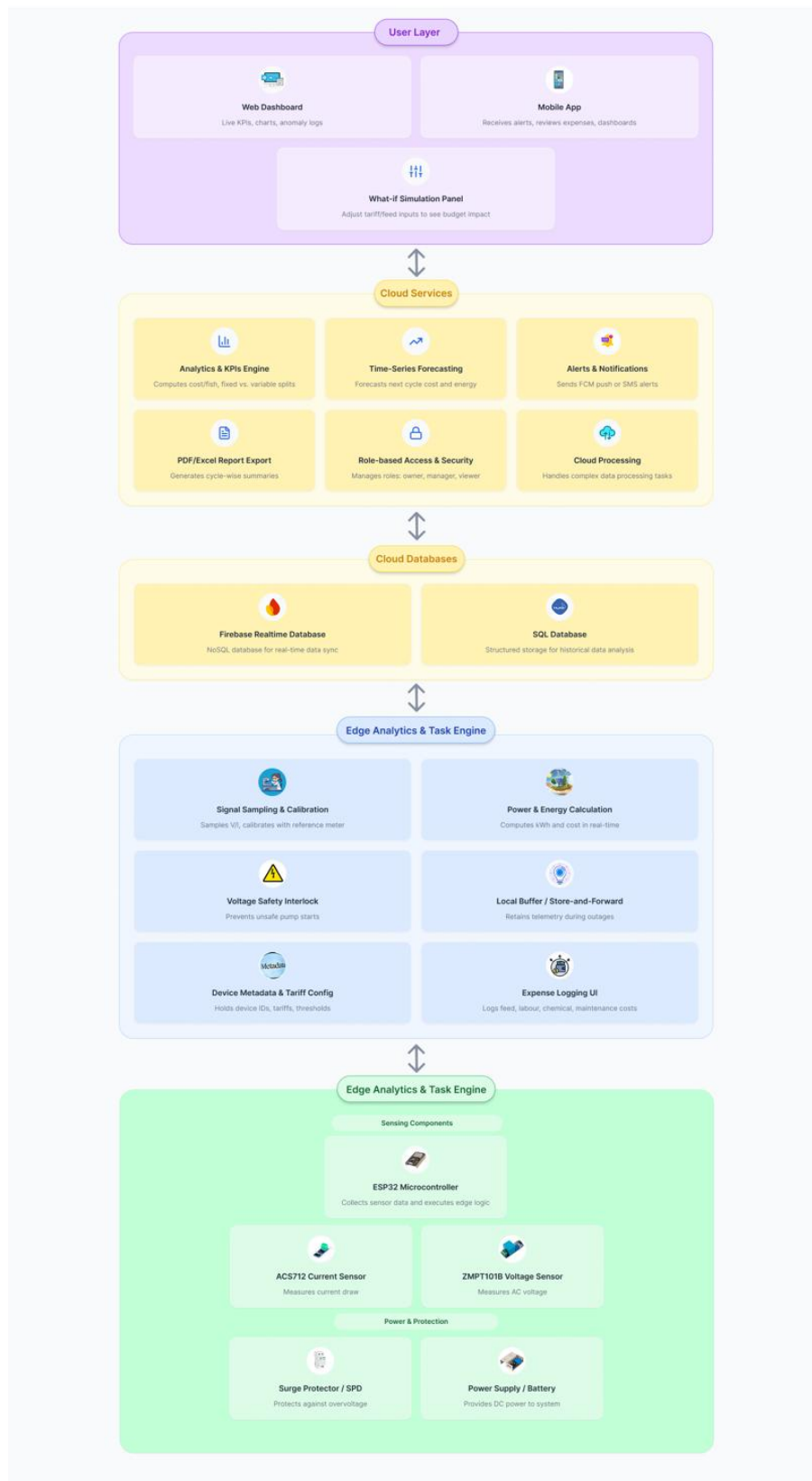


Figure 3.2: component overview

3.3 Gantt chart

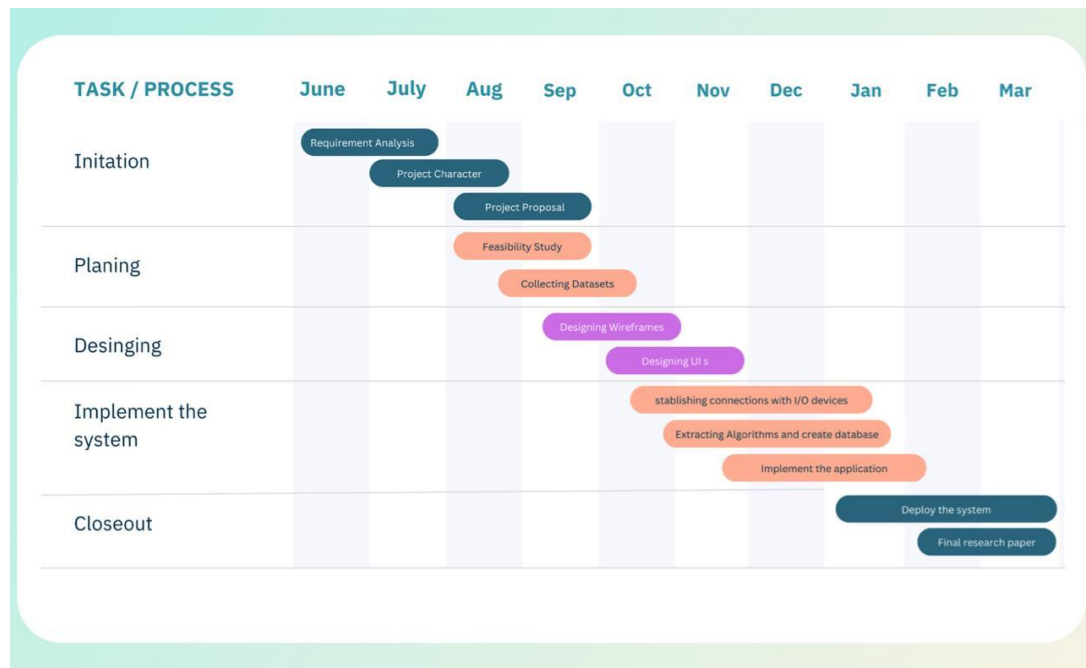


Figure 3.3:Gantt chart

4 PROJECT REQUIREMENTS

4.1 User requirements

- See live site-level power (kW), energy (kWh), and cost for today on your phone or computer.
- In less than 10 seconds, you can enter expenses like feed, fingerlings, chemicals, labour, and maintenance into the POS system. You can also add a category, supplier, unit cost, and notes.
- Look at the cycle KPIs, which are the total spend, the energy share, the category splits, the fixed vs. variable costs, and the cost per fish that survives.
- Get alerts for budget/run-rate problems and unusual consumption (spikes, step changes), as well as voltage-band interlock alerts for pumps.
- You can see the next period's forecasts (in kWh and total cost) with a confidence band and a short summary in plain language.
- Make simple what-if changes (like a 10% increase in tariffs or an 8% increase in feed) and see how they affect KPIs right away.
- You can send cycle reports and dashboards to PDF or Excel for approval and audit.
- Access based on role (owner, manager, viewer) and a record of changes and approvals.

- Offline resilience: data and alerts are stored locally and synced automatically later, so there are no losses or duplicates.
- Setup and calibration are quick: a wizard can match a reference meter with an error of less than 5% in minutes.
- By default, TLS, protected keys, local-time (ISO-8601) timestamps, and configurable retention make it safe.

4.2 Functional requirements

- Measuring energy at the site level: Use ESP32 to measure AC voltage (ZMPT101B/zero-flux) and current (ACS712) at the main feeder or read a smart meter from a utility. Find out the instantaneous power, cumulative energy (kWh), and cost under flat or time-of-use rates.
- Creating a baseline and calibrating: Set sensors to a reference meter, keep track of scale and offset factors, and keep rolling baselines to compare.
- Expense logging UI: Give a POS the ability to keep track of costs for feed, fingerlings, drugs/chemicals, labor, maintenance, and overhead. Make it possible to add tags to entries by pond and production cycle, and let people change notes, suppliers, unit codes, and batches.
- Cycle KPIs and analytics: Find out how much you spend, how much goes to energy, how much is split up by groups and categories, how much is fixed versus variable costs, and how much it costs to keep a fish alive. Keep an eye on the run-rate compared to the budget all the time. Use stock counts and survival to make comparisons fairer by reconciling site-level metering with POS entries to assign costs by pond, cycle, and activity.
- Predicting (cost and energy): Estimate the next period's kWh and total cost, show a clear range of confidence, update at regular intervals, and keep a simple history so people can see how close the forecasts were.
- Voltage protection interlock: Stop pumps or motors from starting outside of a safe voltage range, like $230\text{ V} \pm 10\%$. After a hard fault, require acknowledgment and log return-to-safe events for traceability.
- Alerts and notifications: [11]Send short push notifications (FCM; SMS is optional) with the type of event, the time, the site or pond, and a brief explanation. Use cool-downs to stop getting the same alerts over and over.

- Reports and exports: Make PDF and Excel summaries and reports to get approval and share.
- Offline queue and sync: Store telemetry and alerts during outages and automatically sync when the connection is restored, making sure that no data is lost and events are in the right order.
- Roles and settings: Set up tariffs, safe voltage ranges, device mappings, ponds, cycles, thresholds, recipients, and POS categories. Then, give people access based on their role as an owner, manager, or viewer.
- Integration API: Securely make KPIs, events, and forecasts available to other modules through a simple JSON/REST endpoint.

4.3 Non - functional requirements

- Accuracy: The power or energy is within 5% of what a reference meter says it should be under normal loads. The timestamps are in local time (ISO-8601), and the cost calculations are correct according to tariff rules.
- Quality of the forecast: The first MAE is 15-25%, and it gets better as more data comes in. Keep an eye on the mistake and train again on time.
- Latency: Push alerts are sent within 15 seconds of confirming an event on a normal network; interlock reaction happens within 2 seconds; and dashboard KPIs update almost in real time.
- Availability and reliability: It can run for more than 24 hours without any help from a person; if it fails, the watchdog restarts it; and store-and-forward makes sure that no alerts are lost.
- Performance footprint: The system runs on an ESP32 microcontroller (no GPU), and when it is stable, CPU and RAM usage stays below 70%.
- Usability: It takes less than 10 seconds to enter each expense by hand; the dashboard works on mobile devices; and you can export to PDF or Excel with just one click.
- Privacy and security: TLS for cloud traffic, secrets and keys stored safely, role-based access, and a configurable data-retention window for telemetry and reports.

- Environmental tolerance: The enclosure and mounts can handle rain and splashes (about IP65) and work well in temperatures between 10 and 40 °C and normal outdoor humidity.
- Lightweight scalability: The backend can handle at least 10 edge nodes and many users at once without any delays in notifications.
- Cost ceiling: The students' budget was met by the prototype hardware (ESP32, ZMPT101B, ACS712, ADC or smart meter, relay/interlock, and enclosure).

4.4 Test cases

Test Case ID	Task	Test Objective	Expected Result	Status
TC_01	Measure a 100 W load for 60 minutes.	Verify power, energy, cost, and timestamp accuracy against a reference meter for a steady load	Mean power $\approx$ 100 W; energy $\approx$ 0.10 kWh; both within $\pm 5\%$ of the reference. Tariffed cost computed exactly. Local timestamps recorded in ISO-8601.	Passed
TC_02	Compare readings at 60 W and 500 W.	Check calibration linearity across a practical range and confirm cost conversion over the interval(s).	Each segment's power and energy within $\pm 5\%$ of the reference; cumulative energy equals the sum of segments. Tariffed cost correct for the whole run. No gaps in samples.	Passed
TC_03	Simulate unsafe voltage and start pump.	Confirm voltage-protection interlock and alerting.	Start is blocked in ≤ 2 s. "Unsafe voltage" alert sent to phone in ≤ 15 s. Event logged with fault and restore-to-safe details; resume only when voltage returns to the safe band and is acknowledged.	Passed
TC_04	Add extra load to cause a short current spike.	Verify spike detection, severity tagging, and alert de-duplication.	"High current" anomaly detected in ≤ 15 s. Exactly one alert during the cool-down window. Severity noted. Event written to the log.	Passed

TC_05	Forecast power and cost using past 7 days' data.	Validate forecast sanity, display, and error metrics.	Forecast follows the daily pattern with a reasonable confidence band. Next-day MAE between 15% and 25%. Dashboard shows forecast vs actuals and the error metric.	Passed
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Table 4.1: Test case

5 BUDGET AND JUSTIFICATION

Device	Price (Rs)	Usability
Zeroflux or ZMPT101B Module	1300.00	Voltage monitoring for protection of motors and pumps
ACS712 Current Sensor Module	1450.00	Current measurements for power consumption tracking
ESP 32	1800.00	Micro Controller
SPD OR HRC FUSE, Surge protector	12000.00	For High Voltage Protection
Component Total Budget	16550.00	

Table 5.1: Budget and justification

This list of materials includes the bare minimum hardware needed to safely measure energy and send the data to the analytics and forecasting pipeline. The ZMPT101B voltage sensor and the ACS712 current sensor [12] are the main tools for measuring the power/energy (kWh) and cost of each device. They also make it possible to protect pumps by interlocking the voltage bands. The ESP32 is a low-power microcontroller that takes samples of V/I, does some basic filtering and calibration, and sends the data to the cloud (or gateway) for KPIs, anomaly detection, and time-series forecasting. The SPD or HRC fuse (surge protector) is the most important part of the electrical protection system. It protects motors, the metering circuit, and farm workers from surges and over-voltage, and it also provides a safe fault path, which is very important in outdoor pond environments. These parts work together to provide accurate device-level metering, strong safety, and reliable telemetry at a price that students can afford. This directly supports cost analytics and predictive budgeting.

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7 APPENDIX

